# VALIDATION REPORT
# OPERATIONAL QUALIFICATION
# SPC MODULE

## JR Validated Environment — SPC Module v1.0

| Field | Value |
|---|---|
| Document Number | JR-VR-SPC-001 |
| Title | Validation Report — Operational Qualification, SPC Module |
| Validation Plan | JR-VP-SPC-001 v1.0 |
| System | JR Validated Environment — SPC Module |
| Module Version | 1.0 |
| Document Version | 1.0 |
| Execution Date | 2026-03-18 |
| OQ Result | PASS — 55 / 55 tests passed, 0 failures, 0 deviations |
| Author | Joep Rous |
| Reviewer | [Name] |
| Approver | [Name] |

## Version History

| Version | Date | Author | Description |
|---|---|---|---|
| 1.0 | 2026-03-18 | Joep Rous | Initial OQ execution report. All 55 test cases passed. No deviations. |

## Approval Signatures

| Role | Name | Signature | Date |
|---|---|---|---|
| Author | Joep Rous | | 2026-03-18 |
| Reviewer | [Name] | | |
| Approver (QA) | [Name] | | |

# 1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment SPC Module, version 1.0. It provides objective evidence that all SPC scripts perform as specified in the OQ Validation Plan JR-VP-SPC-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 55 test cases, and the formal OQ conclusion.

# 2. Scope

## 2.1 In Scope

- All 5 R scripts in repos/spc/R/ (jrc_spc_imr, jrc_spc_xbar_r, jrc_spc_xbar_s, jrc_spc_p, jrc_spc_c)
- Correct computation of control limits and all 8 Western Electric rule violations for valid inputs
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- PNG output to ~/Downloads/

## 2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, grid)

# 3. Reference Documents

| Document ID | Title | Location |
|---|---|---|
| JR-VP-SPC-001 v1.0 | Validation Plan — OQ, SPC Module | repos/spc/docs/ignore/spc_validation_plan.docx |
| JR-VP-002 v1.0 | Validation Plan — OQ, Community Script Suite v1.1.0 | docs/ignore/oq_validation_plan.docx |
| JR-IQ-001 | IQ Execution Evidence | docs/IQ_validation_20260311_205146.txt |
| OQ Evidence | SPC OQ Execution Evidence (pytest output + header) | ~/.jrscript/MyProject/validation/spc_oq_execution_<timestamp>.txt |

# 4. Test Environment

| Item | Value |
|---|---|
| Execution date / time | 2026-03-18T19:35:23 |
| Completion date / time | 2026-03-18T19:35:49 |
| Host | Joeps-MacBook-Air.local |
| Operating system | macOS 15.7.3 (Darwin 24.6.0, aarch64) |
| R version | 4.5.0 |
| Python version | 3.11.9 |
| pytest version | 8.3.4 |
| OQ virtual environment | ~/.venvs/MyProject_oq/ (reused from core OQ suite) |
| Script entry point | bin/jrrun (sets RENV_PATHS_ROOT, loads renv library) |

| | |
|---|---|
| Test runner | repos/spc/admin_spc_oq |
| Test directory | repos/spc/oq/ |
| Test data directory | repos/spc/oq/data/ (17 committed synthetic CSV files) |
| Evidence file | ~/.jrscript/MyProject/validation/spc_oq_execution_<timestamp>.txt |

## 5. Test Data

All test data files are synthetic CSV files committed to repos/spc/oq/data/ and included in the project integrity check (admin/project_integrity.sha256). No real patient or production data is used.

| File | Content | Used by |
|---|---|---|
| imr_stable.csv | 25 observations, id/value, mean ≈ 10.07, no WE rule fires | TC-SPC-IMR-001, 002, 005, 006 |
| imr_ooc.csv | Same as stable but obs 13 = 11.50 (Rule 1 violation, UCL ≈ 11.27) | TC-SPC-IMR-003, 004 |
| imr_missing_col.csv | Missing 'value' column | TC-SPC-IMR-009 |
| imr_one_obs.csv | Only 1 observation (< 2 required for moving range) | TC-SPC-IMR-010 |
| xbar_r_stable.csv | 20 subgroups × n = 5, long format; no WE rule fires | TC-SPC-XBR-001, 002, 005, 006 |
| xbar_r_ooc.csv | Same but sg09 replaced with mean ≈ 53 (Rule 1 violation) | TC-SPC-XBR-003, 004 |
| xbar_r_unbalanced.csv | sg01 n = 5, sg02 n = 4, sg03 n = 5 (unbalanced) | TC-SPC-XBR-010, TC-SPC-XBS-010 |
| xbar_r_n_too_large.csv | 3 subgroups × n = 11 (exceeds X-bar/R maximum) | TC-SPC-XBR-011 |
| xbar_r_missing_col.csv | Missing 'value' column | TC-SPC-XBR-009 |
| xbar_s_stable.csv | 20 subgroups × n = 8; three spread types to prevent Rule 7; no WE rule fires | TC-SPC-XBS-001, 002, 005, 006 |
| xbar_s_ooc.csv | Same but sg09 replaced with mean ≈ 102 (Rule 1 violation) | TC-SPC-XBS-003, 004 |
| xbar_s_missing_col.csv | Missing 'value' column | TC-SPC-XBS-009 |
| p_stable.csv | 25 subgroups, n = 100 each, defectives 1–8; p_bar = 0.0416; no WE rule fires | TC-SPC-P-001, 002, 003, 006 |
| p_ooc.csv | Same but subgroup 13 = 15 defectives (p = 0.15 > UCL ≈ 0.10) | TC-SPC-P-004, 005 |
| p_missing_col.csv | Missing 'defectives' column | TC-SPC-P-009 |
| p_invalid.csv | Row with defectives = 101 > n = 100 | TC-SPC-P-010 |
| c_stable.csv | 25 subgroups; defects 1–9; c_bar ≈ 4.96; no WE rule fires | TC-SPC-C-001, 002, 003, 006 |
| c_ooc.csv | Same but subgroup 12 = 15 defects (> UCL ≈ 12.2) | TC-SPC-C-004, 005 |
| c_missing_col.csv | Missing 'defects' column | TC-SPC-C-009 |

## 6. Test Execution Results

### 6.1 Summary

| Item | Value |
|---|---|
| Total test cases | 55 |

| | |
|---|---|
| Passed | 55 |
| Failed | 0 |
| Errors | 0 |
| Duration | 25.59 s |
| OQ result | PASS |

## 6.2 Results by Test File

| Test file | Script(s) covered | Tests | Result |
|---|---|---|---|
| test_spc_imr.py | jrc_spc_imr | 11 | 11 / 11 PASS |
| test_spc_xbar_r.py | jrc_spc_xbar_r | 12 | 12 / 12 PASS |
| test_spc_xbar_s.py | jrc_spc_xbar_s | 11 | 11 / 11 PASS |
| test_spc_p.py | jrc_spc_p | 11 | 11 / 11 PASS |
| test_spc_c.py | jrc_spc_c | 10 | 10 / 10 PASS |

## 6.3 Individual Test Case Results

All 55 test cases passed. The full pytest output, including individual test IDs, pass/fail status, and timing, is recorded in the evidence file:

**~/.jrscript/MyProject/validation/spc_oq_execution_<timestamp>.txt**

That file also contains the evidence header (host, R version, Python version, pytest version, timestamp) and the OQ RESULT: PASS footer. The per-test results are reproduced in the table below for traceability.

| Test Case ID | Description | Expected | Result |
|---|---|---|---|
| TC-SPC-IMR-001 | Happy path — exit 0, sections present | Exit 0, sections present | PASS |
| TC-SPC-IMR-002 | Stable data — IN CONTROL verdict | IN CONTROL or STABLE | PASS |
| TC-SPC-IMR-003 | OOC data — exit 0, OUT OF CONTROL | Exit 0, OUT OF CONTROL/SIGNALS | PASS |
| TC-SPC-IMR-004 | OOC data — obs 13 flagged as Rule 1 | '13' and '[1]' in output | PASS |
| TC-SPC-IMR-005 | --ucl / --lcl flags accepted | Exit 0, limit value in output | PASS |
| TC-SPC-IMR-006 | PNG written to ~/Downloads/ | PNG present, recent mtime | PASS |
| TC-SPC-IMR-007 | No arguments → usage message | Exit ≠ 0, 'Usage' in output | PASS |
| TC-SPC-IMR-008 | File not found | Exit ≠ 0, 'not found' | PASS |
| TC-SPC-IMR-009 | Missing 'value' column | Exit ≠ 0, 'value' in output | PASS |
| TC-SPC-IMR-010 | Too few observations (1 row) | Exit ≠ 0 | PASS |
| TC-SPC-IMR-011 | Bypass protection — direct Rscript call | Exit ≠ 0, RENV_PATHS_ROOT | PASS |
| TC-SPC-XBR-001 | Happy path — exit 0, sections present | Exit 0, sections present | PASS |
| TC-SPC-XBR-002 | Stable data — IN CONTROL verdict | IN CONTROL or STABLE | PASS |
| TC-SPC-XBR-003 | OOC data — exit 0, OUT OF CONTROL | Exit 0, OUT OF CONTROL/SIGNALS | PASS |
| TC-SPC-XBR-004 | OOC data — sg09 flagged | 'sg09' in output | PASS |
| TC-SPC-XBR-005 | --ucl / --lcl flags accepted | Exit 0, limit value in output | PASS |
| TC-SPC-XBR-006 | PNG written to ~/Downloads/ | PNG present, recent mtime | PASS |
| TC-SPC-XBR-007 | No arguments → usage message | Exit ≠ 0, 'Usage' in output | PASS |
| TC-SPC-XBR-008 | File not found | Exit ≠ 0, 'not found' | PASS |
| TC-SPC-XBR-009 | Missing 'value' column | Exit ≠ 0, 'value' in output | PASS |
| TC-SPC-XBR-010 | Unbalanced subgroups | Exit ≠ 0, 'unbalanced' | PASS |

| | | | |
|---|---|---|---|
| TC-SPC-XBR-011 | Subgroup size n > 10 | Exit ≠ 0, suggests xbar_s | PASS |
| TC-SPC-XBR-012 | Bypass protection — direct Rscript call | Exit ≠ 0, RENV_PATHS_ROOT | PASS |
| TC-SPC-XBS-001 | Happy path — exit 0, sections present | Exit 0, sections present | PASS |
| TC-SPC-XBS-002 | Stable data — IN CONTROL verdict | IN CONTROL or STABLE | PASS |
| TC-SPC-XBS-003 | OOC data — exit 0, OUT OF CONTROL | Exit 0, OUT OF CONTROL/SIGNALS | PASS |
| TC-SPC-XBS-004 | OOC data — sg09 flagged | 'sg09' in output | PASS |
| TC-SPC-XBS-005 | --ucl / --lcl flags accepted | Exit 0, limit value in output | PASS |
| TC-SPC-XBS-006 | PNG written to ~/Downloads/ | PNG present, recent mtime | PASS |
| TC-SPC-XBS-007 | No arguments → usage message | Exit ≠ 0, 'Usage' in output | PASS |
| TC-SPC-XBS-008 | File not found | Exit ≠ 0, 'not found' | PASS |
| TC-SPC-XBS-009 | Missing 'value' column | Exit ≠ 0, 'value' in output | PASS |
| TC-SPC-XBS-010 | Unbalanced subgroups | Exit ≠ 0, 'unbalanced' | PASS |
| TC-SPC-XBS-011 | Bypass protection — direct Rscript call | Exit ≠ 0, RENV_PATHS_ROOT | PASS |
| TC-SPC-P-001 | Happy path — exit 0, sections present | Exit 0, sections present | PASS |
| TC-SPC-P-002 | Stable data — IN CONTROL verdict | IN CONTROL or STABLE | PASS |
| TC-SPC-P-003 | Known p-bar ≈ 0.0416 | Output matches 0.04[01234] | PASS |
| TC-SPC-P-004 | OOC data — exit 0, OUT OF CONTROL | Exit 0, OUT OF CONTROL/SIGNALS | PASS |
| TC-SPC-P-005 | OOC data — subgroup 13 flagged | '13' in output | PASS |
| TC-SPC-P-006 | PNG written to ~/Downloads/ | PNG present, recent mtime | PASS |
| TC-SPC-P-007 | No arguments → usage message | Exit ≠ 0, 'Usage' in output | PASS |
| TC-SPC-P-008 | File not found | Exit ≠ 0, 'not found' | PASS |
| TC-SPC-P-009 | Missing 'defectives' column | Exit ≠ 0, 'defective' | PASS |
| TC-SPC-P-010 | Defectives > n | Exit ≠ 0, validation error | PASS |
| TC-SPC-P-011 | Bypass protection — direct Rscript call | Exit ≠ 0, RENV_PATHS_ROOT | PASS |
| TC-SPC-C-001 | Happy path — exit 0, sections present | Exit 0, sections present | PASS |
| TC-SPC-C-002 | Stable data — IN CONTROL verdict | IN CONTROL or STABLE | PASS |
| TC-SPC-C-003 | Known c-bar ≈ 4.96 | Output matches 4.8[0-9]\|4.9[0-9] | PASS |
| TC-SPC-C-004 | OOC data — exit 0, OUT OF CONTROL | Exit 0, OUT OF CONTROL/SIGNALS | PASS |
| TC-SPC-C-005 | OOC data — subgroup 12 flagged | '12' in output | PASS |
| TC-SPC-C-006 | PNG written to ~/Downloads/ | PNG present, recent mtime | PASS |
| TC-SPC-C-007 | No arguments → usage message | Exit ≠ 0, 'Usage' in output | PASS |
| TC-SPC-C-008 | File not found | Exit ≠ 0, 'not found' | PASS |
| TC-SPC-C-009 | Missing 'defects' column | Exit ≠ 0, 'defect' | PASS |
| TC-SPC-C-010 | Bypass protection — direct Rscript call | Exit ≠ 0, RENV_PATHS_ROOT | PASS |

## 7. Deviations from OQ Plan (JR-VP-SPC-001 v1.0)

No deviations from the approved OQ plan were identified during this OQ execution.

| ID | Description | Resolution | Status |
|---|---|---|---|
| — | None | — | N/A |

## 8. Requirements Traceability Matrix

The table below maps each User Requirement from JR-VP-SPC-001 to the test cases executed and their result.

| UR | Requirement (summary) | Test Cases | Result |
|---|---|---|---|
| UR-SPC-001 | jrc_spc_imr — X-bar, MR-bar, control limits | TC-SPC-IMR-001, 002 | 2 / 2 PASS |
| UR-SPC-002 | jrc_spc_imr — WE rules, violation list | TC-SPC-IMR-002, 003, 004 | 3 / 3 PASS |
| UR-SPC-003 | jrc_spc_imr — --ucl / --lcl | TC-SPC-IMR-005 | 1 / 1 PASS |
| UR-SPC-004 | jrc_spc_imr — verdict | TC-SPC-IMR-001, 002, 003 | 3 / 3 PASS |
| UR-SPC-005 | jrc_spc_imr — PNG to ~/Downloads/ | TC-SPC-IMR-006 | 1 / 1 PASS |
| UR-SPC-006 | jrc_spc_imr — reject invalid inputs | TC-SPC-IMR-007, 008, 009, 010 | 4 / 4 PASS |
| UR-SPC-007 | jrc_spc_xbar_r — grand mean, R-bar, limits | TC-SPC-XBR-001, 002 | 2 / 2 PASS |
| UR-SPC-008 | jrc_spc_xbar_r — WE rules, violation list | TC-SPC-XBR-002, 003, 004 | 3 / 3 PASS |
| UR-SPC-009 | jrc_spc_xbar_r — --ucl / --lcl | TC-SPC-XBR-005 | 1 / 1 PASS |
| UR-SPC-010 | jrc_spc_xbar_r — verdict | TC-SPC-XBR-001, 002, 003 | 3 / 3 PASS |
| UR-SPC-011 | jrc_spc_xbar_r — PNG to ~/Downloads/ | TC-SPC-XBR-006 | 1 / 1 PASS |
| UR-SPC-012 | jrc_spc_xbar_r — reject unbalanced / n > 10 | TC-SPC-XBR-010, 011 | 2 / 2 PASS |
| UR-SPC-013 | jrc_spc_xbar_s — grand mean, S-bar, limits via c4 | TC-SPC-XBS-001, 002 | 2 / 2 PASS |
| UR-SPC-014 | jrc_spc_xbar_s — WE rules, violation list | TC-SPC-XBS-002, 003, 004 | 3 / 3 PASS |
| UR-SPC-015 | jrc_spc_xbar_s — --ucl / --lcl | TC-SPC-XBS-005 | 1 / 1 PASS |
| UR-SPC-016 | jrc_spc_xbar_s — verdict | TC-SPC-XBS-001, 002, 003 | 3 / 3 PASS |
| UR-SPC-017 | jrc_spc_xbar_s — PNG to ~/Downloads/ | TC-SPC-XBS-006 | 1 / 1 PASS |
| UR-SPC-018 | jrc_spc_xbar_s — reject unbalanced subgroups | TC-SPC-XBS-010 | 1 / 1 PASS |
| UR-SPC-019 | jrc_spc_p — p-bar, variable limits, WE rules | TC-SPC-P-001, 002, 003, 004, 005 | 5 / 5 PASS |
| UR-SPC-020 | jrc_spc_p — verdict | TC-SPC-P-001, 002, 004 | 3 / 3 PASS |
| UR-SPC-021 | jrc_spc_p — reject defectives > n | TC-SPC-P-010 | 1 / 1 PASS |
| UR-SPC-022 | jrc_spc_p — PNG to ~/Downloads/ | TC-SPC-P-006 | 1 / 1 PASS |
| UR-SPC-023 | jrc_spc_c — c-bar, sigma, limits, WE rules | TC-SPC-C-001, 002, 003, 004, 005 | 5 / 5 PASS |
| UR-SPC-024 | jrc_spc_c — verdict; PNG to ~/Downloads/ | TC-SPC-C-001, 002, 004, 006 | 4 / 4 PASS |
| UR-SPC-025 | All scripts — RENV_PATHS_ROOT bypass protection | TC-SPC-IMR-011, XBR-012, XBS-011, P-011, C-010 | 5 / 5 PASS |
| UR-SPC-026 | All scripts — no-arguments usage message | TC-SPC-IMR-007, XBR-007, XBS-007, P-007, C-007 | 5 / 5 PASS |
| UR-SPC-027 | All scripts — file not found error | TC-SPC-IMR-008, XBR-008, XBS-008, P-008, C-008 | 5 / 5 PASS |

## 9. OQ Conclusion

The Operational Qualification of the JR Validated Environment SPC Module v1.0 is complete. All 55 test cases defined in JR-VP-SPC-001 were executed on 2026-03-18 and passed with no failures, no errors, and no deviations.

All 27 user requirements (UR-SPC-001 through UR-SPC-027) are covered by at least one passing test case. The OQ acceptance criteria stated in JR-VP-SPC-001 Section 7 are fully met.

**The SPC module is hereby declared OPERATIONALLY QUALIFIED and released for use in process monitoring and control activities in accordance with the JR Validated Environment quality system.**

*OQ Evidence file: ~/.jrscript/MyProject/validation/spc_oq_execution_<timestamp>.txt*